



USER GUIDE

F1 DATABASE

Practical: Thursday 9-12

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Creating account and logging in:

- Enter: CREATE USER 'me'@'localhost' IDENTIFIED BY 'Password123';
- Enter: mysql -u me -p
- Enter the password

Creating and opening the database:

- Enter: CREATE DATABASE F1_21174229;
- Enter: USE F1_21174229;

To create the tables and view structure:

- Enter: SOURCE Tables.sql;
- You can use SHOW and DESC commands to see the tables and their attributes.
- SHOW TABLES;
- DESC (Table_Name). Example: DESC Drivers;

```
nihalz132@UbuntuDesktop2:~/Desktop/DBSFinal$ mysql -u me -p
Enter password:
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 42
Server version: 8.0.41-0ubuntu0.24.10.1 (Ubuntu)

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owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> USE F1_21174229;
Reading table information for completion of table and column names
You can turn off this feature to get a quicker startup with -A

Database changed
mysql> SOURCE Tables.sql;
Query OK, 0 rows affected (0.04 sec)

Query OK, 0 rows affected (0.06 sec)

Query OK, 0 rows affected (0.05 sec)

Query OK, 0 rows affected (0.05 sec)

Query OK, 0 rows affected (0.03 sec)

Query OK, 0 rows affected (0.04 sec)
```

To add data and check the data in a table:

- Enter: SOURCE Values.sql;
- You can also use SELECT to see the data in each table.
- Example: SELECT * FROM Teams;

```
mysql> SOURCE Values.sql;
Query OK, 12 rows affected (0.02 sec)
Records: 12  Duplicates: 0  Warnings: 0

Query OK, 3 rows affected (0.02 sec)
Records: 3  Duplicates: 0  Warnings: 0

Query OK, 30 rows affected (0.02 sec)
Records: 30  Duplicates: 0  Warnings: 0

Query OK, 25 rows affected (0.02 sec)
Records: 25  Duplicates: 0  Warnings: 0

Query OK, 8 rows affected (0.02 sec)
Records: 8  Duplicates: 0  Warnings: 0

Query OK, 68 rows affected (0.03 sec)
Records: 68  Duplicates: 0  Warnings: 0

Query OK, 30 rows affected (0.01 sec)
Records: 30  Duplicates: 0  Warnings: 0

Query OK, 176 rows affected (0.03 sec)
Records: 176  Duplicates: 0  Warnings: 0

Query OK, 1359 rows affected (0.16 sec)
Records: 1359  Duplicates: 0  Warnings: 0
```

Checking the queries:

- Enter: SOURCE Queries.sql;
- This will output all the 8 queries that were implemented in the file

```
mysql> SOURCE Queries.sql;
+-----+-----+
| Track  | Location |
+-----+-----+
| Bahrain | Sakhir   |
| Belgium | Stavelot |
| Brazil  | São Paulo |
+-----+-----+
3 rows in set (0.00 sec)

+-----+-----+
| Year | Circuit_ID |
+-----+-----+
| 2023 | C01        |
| 2023 | C02        |
| 2023 | C03        |
| 2023 | C04        |
| 2023 | C06        |
| 2023 | C08        |
| 2023 | C09        |
| 2023 | C10        |
| 2023 | C11        |
| 2023 | C12        |
| 2023 | C13        |
| 2023 | C14        |
+-----+-----+
```

Checking the implemented advanced concepts:

- Enter: SOURCE Advanced_Concepts.sql;
- Use call() for stored procedures.
- Use to see available procedures: SHOW PROCEDURE STATUS WHERE D = 'F1_21174229';
- **Procedure 1:** Call Add_Driver(Driver_ID, Driver_Number, Driver_Name, Team_Name, DOB, Nationality)
- **Procedure 2:** CALL Get_Driver_Race_Count(Driver_ID, @Race_Count);
SELECT @Race_Count;
- **Procedure 3:** CALL Get_Race_Winners_By_Year(Year);

```
mysql> CALL Add_Driver('F1031', 33, 'Max Verstappen', 'Red Bull Racing', '1997-09-30', 'Dutch');
Query OK, 1 row affected (0.03 sec)
```

```
mysql> CALL Get_Driver_Race_Count('F1001', @Race_Count);
Query OK, 1 row affected (0.00 sec)
```

```
mysql> SELECT @Race_Count;
+-----+
| @Race_Count |
+-----+
|          68 |
+-----+
1 row in set (0.01 sec)
```

```
mysql> CALL Get_Race_Winners_By_Year(2023);
```

```
+-----+-----+
| Track      | Driver_Name |
+-----+-----+
| Bahrain    | Max Verstappen |
| Saudi Arabia | Sergio Perez |
| Australia  | Max Verstappen |
| Japan      | Max Verstappen |
| Miami      | Max Verstappen |
| Monaco     | Max Verstappen |
| Canada     | Max Verstappen |
| Spain      | Max Verstappen |
| Austria    | Max Verstappen |
| Great Britain | Max Verstappen |
| Hungary    | Max Verstappen |
```

- **View 1:** SELECT * FROM Most_Points_Top10;
- **View 2:** SELECT * FROM Most_Wins;

```
mysql> SELECT * FROM Most_Points_Top10;
+-----+-----+-----+
| Position | Driver_Name | Total_Points |
+-----+-----+-----+
| 1 | Max Verstappen | 1362 |
| 2 | Charles Leclerc | 803 |
| 3 | Sergio Perez | 689 |
| 4 | Carlos Sainz | 668 |
| 5 | Lewis Hamilton | 657 |
| 6 | George Russell | 645 |
| 7 | Lando Norris | 644 |
| 8 | Fernando Alonso | 349 |
| 9 | Oscar Piastri | 347 |
| 10 | Esteban Ocon | 168 |
+-----+-----+-----+
10 rows in set (0.01 sec)
```

```
mysql> SELECT * FROM Most_Wins;
+-----+-----+-----+
| Driver_ID | Driver_Name | Total_Wins |
+-----+-----+-----+
| F1001 | Max Verstappen | 43 |
| F1002 | Charles Leclerc | 6 |
| F1003 | Sergio Perez | 4 |
| F1005 | Carlos Sainz | 4 |
| F1007 | Lando Norris | 4 |
| F1004 | George Russell | 3 |
| F1006 | Lewis Hamilton | 2 |
| F1024 | Oscar Piastri | 2 |
+-----+-----+-----+
```

Connecting to python and running the queries:

- Now exit from mysql, type: exit
- Enter: pip install mysql-connector-python
- For defined queries: python3 Defined_Queries.py

```
(myenv) nihalz132@UbuntuDesktop2:~/Desktop/DBSFinal$ python Defined_Queries.py
```

```
Query 1: Top 10 Drivers Standings 2024:
(2024, 1, 'F1001', 'Red Bull Racing Honda RBPT', 437)
(2024, 2, 'F1007', 'McLaren Mercedes', 374)
(2024, 3, 'F1002', 'Ferrari', 356)
(2024, 4, 'F1024', 'McLaren Mercedes', 292)
(2024, 5, 'F1005', 'Ferrari', 290)
(2024, 6, 'F1004', 'Mercedes', 245)
(2024, 7, 'F1006', 'Mercedes', 223)
(2024, 8, 'F1003', 'Red Bull Racing Honda RBPT', 152)
(2024, 9, 'F1009', 'Aston Martin Aramco Mercedes', 70)
(2024, 10, 'F1014', 'Alpine Renault', 42)
```

```
Query 2: Top 5 Teams with the Most Championship Points:
('Red Bull Racing', 2208)
('Ferrari', 1612)
('Mercedes', 1392)
('McLaren', 1127)
('Aston Martin', 429)
```

```
Query 3: Top 5 Most Expensive Tickets:
(8, 'Abu Dhabi', 'Paddock Club', 7700.0)
(8, 'Monaco', 'Paddock Club', 7500.0)
(8, 'Miami', 'Paddock Club', 6500.0)
(8, 'Great Britain', 'Paddock Club', 5300.0)
(8, 'United States', 'Paddock Club', 5300.0)
```

Running the interaction menu:

- Enter: python3 Interaction_Menu.py
- Select any option from the menu and add/edit/delete the data
- To exit, enter 8

```
(myenv) nihalz132@UbuntuDesktop2:~/Desktop/DBSFinal$ python Interaction_Menu.py
FORMULA 1 DB MANAGEMENT MENU:
1. Insert Team
2. Insert Driver
3. Insert Race Result
4. Update Team
5. Update Driver
6. Delete Team
7. Delete Driver
8. Exit
Enter your choice: 6
Enter team name to delete: curtin
Error: Team not found.
```